Projekti – OR 2026/2027
pri predmetu Finančni praktikum*

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1 Osnovno o projektu

Osnove:

- Pričakovana obremenitev posameznega študenta za projekt je 60 ur.
- Vsak projekt izdela skupina treh študentov. Skupine so naključne in jih določi učitelj.
- Študenti v isti skupini lahko dobijo različne ocene.
- Vsaka skupina dobi smernico. Pri določanju točne vsebine imate nekaj **fleksibilnosti**.
- Projekt mora vsebovati programiranje in **eksperimente**. Lahko si izberete programsko okolje, v katerem boste delali. Lahko najdete podatke na internetu, lahko jih pa tudi generirate sami.
- Na koncu projekta bo zagovor.
- Asistent A. Požun in prof. R. Škrekovski sva na voljo na konzultacije, pomoč itd.

Predlagamo, da svoje programe in poročila sproti objavljate na **GitHub** ali podobno. Priporočamo tudi, da za programiranje uporabljate okolje **Sage** (SageMath) – gre za okolje, osnovano na programskem jeziku Python, z dodano podporo za matematiko (posebej bo prišla v upoštev podpora za grafe in celoštevilsko linearno programiranje). Kratek opis okolja Sage, generatorja grafov `geng`, celoštevilskega linearnega programiranja, SAT reševalnikov, hevrističnih metod (random walk, hill climbing, simulated annealing) in dela z multigrafi bo v ločenem dokumentu *Computational tools for the projects*.

Na koncu bomo imeli sestanek oziroma zagovor projekta s predavateljem in asistentom, kjer bo določena ocena projekta. Za zagovor, ki bo trajal do pol ure, lahko pripravite kratko predstavitev (max 20 min). Na zagovoru pričakujemo tudi argumentacijo smiselnosti vašega dela (zakaj ste se tako lotili) ter seveda razumevanje kode programa in eksperimentov.

Naloge:

- Razumevanje problema.
- Izbira natančnega problema (ali svoje variante problema) iz reference.
- Domneve v projektih so odprte: tudi potrditev domneve za majhne primere ali negativen rezultat (npr. da protiprimera niste našli) je veljaven rezultat.

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- Izdelava kratkega (do 2 strani) opisa problema in načrta za nadaljnje delo.
- Programiranje rešitev ali rešitve izbranega problema.
- Generiranje naključnih ali iskanje realnih podatkov za problem.
- Eksperimentiranje z napisanim programom in generiranimi podatki.
- Poročilo.
- Za naprogramirano kodo je potrebno pripraviti navodila za uporabo (lahko tudi kot komentar v kodi). Koda programa mora biti lepo komentirana.
- Zagovor, kjer predstavite izdelek. Pomembno je, da razumete in lahko zagovarjate, kaj ste delali in uporabljali (tudi kodo programa).

Poročilo:

- Približno 5 strani (fleksibilno).
- Vsebuje jasen opis problema.
- Vsebuje glavne ideje programa ali psevdokodo.
- Vsebuje jasen opis generiranja ali izbiranja podatkov.
- Prikaže grafe, tabele ali drugačen opis eksperimentov (npr. kako se poveča čas izvajanja z velikostjo problema).

2 Koledar

Letos bomo imeli dva termina za projekt. Termine ste si že izbrali na spletni učilnici. Če si jih kdo še ni izbral, naj to stori do 21. oktobra 2026. Z ostalimi iz istega termina ste bili naključno razdeljeni v skupine, vsaka skupina pa je dobila svoj projekt.

Pri nekaterih projektih bo morda kaka skupina v drugem terminu nadaljujevala delo skupine iz prvega termina in bo lahko uporabljala njeno kodo in rezultate. Ker se drugi termin začne pred rokom za oddajo prvega, prosimo skupine prvega termina, da svojo kodo sproti objavljajo (npr. na GitHubu).

Prvi termin:

- Dodelitev skupin in tem: do 23. oktober 2026.
- Rok za kratek opis: 13. november 2026.
- Rok za poročilo in program: 11. december 2026.
- Zagovori: 14.–18. december 2026. Datumi bodo določeni kasneje.

Drugi termin:

- Dodelitev skupin in tem: 4. december 2026.
- Rok za kratek opis: 8. januar 2027.
- Rok za poročilo in program: 29. januar 2027.
- Zagovori: 1.–5. februar 2027. Datumi bodo določeni kasneje.

3 Projekti

3.1 Metric dimension of directed circulant graphs

For a directed graph G and vertices u, v, let $d(u,v)$ be the length of a shortest directed path from u to v. For an ordered set $W = \{w_1, \ldots, w_z\}$ of vertices, let $r(W|v) = (d(w_1,v), \ldots, d(w_z,v))$. The set W is *resolving* if the vectors $r(W|v)$ are pairwise distinct, and the *metric dimension* $\dim(G)$ is the size of a smallest resolving set. For generators $1 \le a_1 < a_2 < \cdots < a_k \le n-1$, the *directed circulant graph* $C_n(a_1, \ldots, a_k)$ has vertices $v_0, \ldots, v_{n-1}$ and arcs $v_i v_{i+a_j}$ for $0 \le i \le n-1$, $1 \le j \le k$ (indices modulo n). It is strongly connected, so that all distances are finite, if and only if $\gcd(n, a_1, \ldots, a_k) = 1$.

The metric dimension of directed circulants with two generators was studied by Vetrík [22]. For consecutive generators it is known [19] that $\dim(C_n(1,2,\ldots,k)) = k$ for $n > 2(k-1)^2$, and that $\dim(C_n(1,\ldots,n-1)) = n-1$ and, for $n \ge 5$, $\dim(C_n(1,\ldots,n-2)) = \dim(C_n(1,\ldots,n-3)) = \lfloor n/2 \rfloor$. All values for $n \le 40$ were computed in the student project [21] (ask us for it; you may reuse its ILP), and the threshold for $\dim = k$ was studied in [20]. Notice that in these reports $C(n,d)$ denotes $C_n(1,2,\ldots,d)$, i.e. there d is the number of generators, while below d is the number of missing generators.

In our work in progress we study the graphs $C_n(1,2,\ldots,n-d)$ for $d \ge 4$. So far,

$$\dim(C_n(1,\ldots,n-4)) = \left\lceil \tfrac{2n}{5} \right\rceil \ (n \ge 10), \qquad \dim(C_n(1,\ldots,n-5)) = \begin{cases} \lfloor n/3 \rfloor & \text{if } 3 \mid n, \\ \lfloor n/3 \rfloor + 2 & \text{if } 3 \nmid n, \end{cases} \quad (n \ge 11),$$

and we propose the following conjecture.

Conjecture 1. *Let $d \ge 4$ and $n > 2d$. Then*

$$\dim(C_n(1,\ldots,n-d)) \ge \begin{cases} \left\lfloor \frac{n}{d+1} \right\rfloor \left\lceil \frac{d+1}{3} \right\rceil & \text{if } d \not\equiv 0 \pmod 3, \\ \frac{n}{2d-1} \cdot \frac{2d}{3} & \text{if } d \equiv 0 \pmod 3 \end{cases}$$

(since the dimension is an integer, the bound can be rounded up). For $d \not\equiv 0 \pmod 3$ equality holds if and only if $(d+1) \mid n$, and for $d \equiv 0 \pmod 3$ if and only if $(2d-1) \mid n$.

Notice that for $d < n/2$ every vertex reaches every other vertex in at most two steps: $d(v_i, v_j) = 1$ if $j - i \pmod n \in \{1, \ldots, n-d\}$ and $d(v_i, v_j) = 2$ otherwise ($j \ne i$). So deciding whether a set is resolving is a purely combinatorial question, and the ILP is simple and fast. If you do not have access to the papers or reports, contact us and we will send them to you.

Project 1: Metric dimension of $C_n(1,2,\ldots,n-d)$: exact values

1. Implement a program (e.g. an ILP) that computes $\dim(C_n(1,\ldots,n-d))$ and an optimal resolving set. You may write the code on your own, or use AI tools to help you design and write it. Check it against the known values above.

2. Compute $\dim(C_n(1,\ldots,n-d))$ for $4 \le d \le 15$ and all n with $2d < n$, going as far beyond $n = 40$ as possible (at least $n = 100$ for $5 \le d \le 10$). For $d = 4, 5$ the values are known, so use them mainly as a test of your program. Notice that for some parameters (e.g. $d = 4$ and $n \ge 90$) the ILP solver of Sage becomes slow; it helps to break the rotational symmetry by fixing $v_0 \in W$, or to try another solver.

3. Test Conjecture 1, including the characterization of the equality cases.

4. For $d = 6, 7, 8, \ldots$ try to guess exact formulas for $\dim(C_n(1, \ldots, n-d))$, e.g. depending on n modulo $d+1$ or modulo $2d-1$. For which n does each formula start to hold?

Report your results.

3.2 Vertex coloring

Odd colorings. An *odd coloring* of a graph G is a proper vertex coloring of G such that every non-isolated vertex v has a color which appears an odd number of times in its neighborhood $N(v)$. The minimum number of colors in an odd coloring of G is the *odd chromatic number* $\chi_o(G)$. Odd colorings were introduced in [28], where it was proved that $\chi_o(G) \leq 9$ for every planar graph G and conjectured that $\chi_o(G) \leq 5$ for every planar graph. The best known general bound for planar graphs is 8 [27]. The conjectured bound would be best possible, since $\chi_o(C_5) = 5$; more generally, $\chi_o(C_n) = 3$ if n is divisible by 3, $\chi_o(C_5) = 5$, and $\chi_o(C_n) = 4$ otherwise.

A *k-cycle* is a cycle of length k. Two cycles of a graph are *adjacent* if they share at least one edge, and *intersecting* if they share at least one vertex. So intersecting cycles need not be adjacent (e.g. two triangles with one common vertex). A natural question is whether planar graphs in which short cycles are "far apart" need fewer colors. Notice that C_5 contains only one cycle, so it belongs to every such class. It is not the only such graph: for example, two 5-cycles with one common vertex also need 5 colors (check it!). Therefore the question is whether every graph in the class which needs 5 colors is built from 5-cycles in a simple way, or whether there are infinitely many essentially different ones.

Project 2: Odd colorings of planar graphs with no adjacent small cycles

1. Implement a program that computes $\chi_o(G)$ (backtracking is recommended; in an ILP the parity condition needs auxiliary integer variables), and check it on cycles. You may write the code on your own, or use AI tools to help you design and write it.

2. Using `geng` with the planarity filter `planarg` (both are part of nauty, included in Sage), or `plantri` (which generates plane embeddings, so the same graph may appear several times), compute $\chi_o(G)$ for all connected planar graphs of small order in each of the following classes:

   (a) planar graphs with no two adjacent 3-cycles;

   (b) planar graphs with no two adjacent cycles of length at most 4 (resp. at most 5);

   (c) planar graphs with no two intersecting cycles of length at most 4;

   (d) planar graphs with no 4-cycle adjacent to two or more 4-cycles;

   (e) planar graphs with no two intersecting cycles of length at most 5.

   For each class, record the largest value of $\chi_o(G)$ for every order, and list all graphs which need 5 colors (as far as possible, at least up to 10 vertices; since the graphs in these classes are sparse, you can bound the number of edges in `geng` to speed up the generation). For which classes is C_5 the only one? Describe the structure of the graphs which need 5 colors: do they all contain 5-cycles? Is there an infinite family?

3. Can the number of colors be even smaller? Find classes of the above type in which every graph other than C_5 seems to have $\chi_o(G) \leq 4$. Is there a natural class of planar graphs with $\chi_o(G) \leq 3$? (Notice that already the cycles C_n with $3 \nmid n$ need 4 colors.) For known results on sparse planar graphs see [26, 25]; e.g. every planar graph of girth at least 11 has $\chi_o(G) \leq 4$, and every planar graph of girth at least 5 has $\chi_o(G) \leq 6$ [25]. Formulate your conjectures

and test them on larger graphs: generate random planar graphs in the class (e.g. delete edges of a random planar graph until the condition holds), look for an odd 4-coloring by backtracking or simulated annealing, and verify every suspected counterexample with your exact program (a heuristic gives only upper bounds).

Report your results.

Conflict-free colorings. A coloring of the vertices of a graph G (not necessarily proper) is *conflict-free with respect to closed neighborhoods* if for every vertex v some color appears exactly once in the closed neighborhood $N[v] = N(v) \cup \{v\}$. The minimum number of colors of such a coloring is denoted by $\chi_{\text{iCFc}}(G)$. Every proper coloring is conflict-free (each vertex sees its own color exactly once), so $\chi_{\text{iCFc}}(G) \leq \chi(G)$, and by the Four Color Theorem $\chi_{\text{iCFc}}(G) \leq 4$ for every planar graph G (Abel et al. [1] showed that three colors suffice if some vertices may remain uncolored, which is a different setting). Fabrici, Lužar, Rindošová and Soták [2] conjectured that 3 colors always suffice for planar graphs. With the help of AI, we disproved the conjecture in [4]: we constructed planar graphs with $\chi_{\text{iCFc}}(G) = 4$, obtained from a planar graph with chromatic number 4 by attaching to each of its edges xy two copies of a small gadget (a 4-cycle joined to x by four edges and to y by one edge); for K_4 this gives 52 vertices. Then a computer search found such a graph on 25 vertices and 55 edges, consisting of K_4 and five small gadgets with $5, 5, 5, 4$ and 2 new vertices (two of them attached to the same edge of K_4, and the smallest one attached to three vertices of K_4). In graph6 format:

```
X~`EERMg@?C?_CgFG?A??O?LO?u??A??B???_??e??FK???K??@
```

An exhaustive search also shows that every planar graph on at most 12 vertices has $\chi_{\text{iCFc}}(G) \leq 3$. Hence the smallest planar graph with $\chi_{\text{iCFc}}(G) = 4$ has between 13 and 25 vertices. If you do not have access to the papers, contact us and we will send them to you.

The work is split into two groups – group A uses exact methods, and group B searches heuristically for a smaller counterexample. In both groups, the main goal is to find a planar graph with $\chi_{\text{iCFc}}(G) = 4$ on fewer than 25 vertices, or to give evidence that no such graph exists on few vertices.

Project 3: Conflict-free colorings – A: exact methods and gadgets

1. Implement a program that decides whether $\chi_{\text{iCFc}}(G) \leq 3$, using a SAT solver (recommended; see the section on SAT solvers in the document *Computational tools for the projects*), an ILP or backtracking. You may write the code on your own, or use AI tools to help you design and write it. Verify that the above graph on 25 vertices has $\chi_{\text{iCFc}}(G) = 4$. Implement also a function computing the minimum number of vertices with no unique color in $N[v]$ over all 3-colorings, e.g. by SAT with a cardinality constraint "at most t such vertices" for $t = 0, 1, 2, \ldots$, or by MaxSAT (an ILP is much slower); it is used by group B.

2. Extend the exhaustive search to 13 vertices for suitable subclasses of planar graphs (use `plantri`, available in Sage as `graphs.planar_graphs`, see the document *Computational tools for the projects*). For 3-connected planar graphs this is feasible (about 10^8 graphs, `graphs.planar_graphs(13, minimum_connectivity=3)`): first try a fast greedy coloring and call SAT only for the remaining graphs (expect a run of about a day; run it in the background, or split it over several computers). Notice that the known example has six vertices of degree 2 and is only 2-connected; however, all 2-connected planar graphs of order 13 are out of reach (about 10^{10} plane graphs), and `plantri` cannot filter them by the number of vertices of degree 2. Build smaller classes differently: e.g. take 3-connected planar graphs (or K_4 and odd wheels with a few extra vertices) on $13 - k$ vertices and

add k vertices of degree 2, each joined to two vertices of a common face. (A minimal counterexample has no vertices of degree 1.)

3. Search for smaller gadgets. Call a planar graph H with an edge xy a *forcing gadget* if there is no 3-coloring of H with $c(x) = c(y)$ in which every vertex of H other than x and y has a unique color in its closed neighborhood, and some color appears at most once in $N_H[x]$ and some color appears at most once in $N_H[y]$. (If forcing gadgets are attached to all edges of a planar graph G_0 with $\chi(G_0) = 4$, every conflict-free 3-coloring of the result is proper on G_0, so $\chi_{\text{iCFc}} = 4$.) In the construction of [4], two copies of the gadget above attached to an edge xy form a forcing gadget with 8 vertices besides x and y. Search systematically (by SAT) all small forcing gadgets that can be drawn in a face incident with xy, and find the smallest ones. Consider also gadgets attached to three or more vertices, as in the 25-vertex graph. Then attach them to K_4 or to other small planar graphs with chromatic number 4 (e.g. odd wheels) and compute the order of the resulting graphs.

4. Shrink the known example: delete vertices and edges of the 25-vertex graph (and of the graphs from task 3) as long as $\chi_{\text{iCFc}}(G) = 4$ is preserved, and find all minimal graphs obtained in this way.

A negative result (e.g. "there is no 3-connected counterexample on 13 vertices") is also a valid result. Report your results.

3.3 Rich edge-colorings

To obtain an edge coloring of a graph G of maximum degree Δ, we assign colors (actually numbers $1, 2, \dots$) to edges so that every pair of adjacent edges (i.e., sharing a common vertex) has different colors. For such a coloring, we need at least Δ colors.

Let G be a graph with a proper edge-coloring. An edge e is called *rich* if all edges adjacent to e receive pairwise distinct colors. A proper edge-coloring is called a *rich edge-coloring* if every non-isolated vertex is incident with at least one rich edge. The minimum number of colors required for such a coloring is called the *rich chromatic index* of G and is denoted by $\chi'_r(G)$.

A graph is called *subcubic* if its maximum degree is at most 3. Five colors do not suffice for all subcubic graphs: the graph H_0 obtained from K_4 by subdividing one edge once has $\chi'_r(H_0) = 6$. Deng, Zhou, and Zhu proposed the following conjecture.

Conjecture 2. *For every connected subcubic graph G other than H_0,*

$$\chi'_r(G) \leq 5.$$

The conjecture is known to hold if G has no adjacent vertices of degree 3, or if G is K_4-minor-free. Moreover, $\chi'_r(K_4) = 5$, so the proposed upper bound would be best possible.

Reference: [8]; see Conjecture 1.2 and Theorem 1.3. If you do not have access to the paper, contact us and we will send it to you.

The work is split into four groups – group A deals with small graphs systematically, group B with medium and large graphs, group C with graphs of larger maximum degree, and group D with multigraphs.

Project 4: Rich edge-colorings – A: small graphs

1. Implement a program that decides whether a given graph admits a rich edge-coloring with at most k colors (e.g. by ILP, SAT, or backtracking), and returns an explicit coloring whenever one exists. If you wish, use AI tools to help you design and write the code. Since

an ILP is already given in the document *Computational tools for the projects*, implement at least two methods (e.g. ILP and SAT) and compare their running times as a function of the order.

2. Test Conjecture 2 for various classes of graphs, e.g. prisms $C_n \square K_2$, Möbius ladders (the circulants $C_{2n}(1, n)$), generalized Petersen graphs $P(n, k)$, planar cubic graphs (from `geng -c -d3 -D3` with a planarity test), fullerenes and snarks (e.g. from the House of Graphs), for as large graphs as possible.

3. Test Conjecture 2 on all connected subcubic graphs of small order (use `geng`), as far as it goes. For every order n, record the largest value of $\chi'_r(G)$, the number of graphs that require five colors (there are already thousands of them for $n = 12$), and list all graphs that require six or more colors.

Report your results.

3.4 Rich-neighbor edge-colorings

Let G be a graph with a proper edge-coloring, and recall that an edge e is *rich* if all edges adjacent to e receive pairwise distinct colors. The coloring is a *rich-neighbor edge-coloring* if every non-isolated edge (i.e. every edge adjacent to some other edge) is adjacent to at least one rich edge. The minimum number of colors of such a coloring is the *rich-neighbor chromatic index* $\chi'_{\mathrm{rn}}(G)$. In a strong edge-coloring every edge is rich, so $\chi'_{\mathrm{rn}}(G) \leq \chi'_s(G)$. Notice that if G is a Δ-regular (simple) graph, a rich edge and its $2\Delta - 2$ adjacent edges receive pairwise distinct colors, so $\chi'_{\mathrm{rn}}(G) \geq 2\Delta - 1$.

This coloring was introduced in [29], where it was proved that $\chi'_{\mathrm{rn}}(G) \leq 7$ for every subcubic multigraph G, and conjectured that $\chi'_{\mathrm{rn}}(G) \leq 5$ for every connected subcubic multigraph $G \neq K_4$ (while $\chi'_{\mathrm{rn}}(K_4) = 6$).

A connected graph is called *exceptional* if it is one of the five graphs C_4, C_7, K_4, $K_{2,2,2}$ and K_6. Each of them is regular and has $\chi'_{\mathrm{rn}}(G) = 2\Delta$. The following conjecture was proposed in [23].

Conjecture 3. *Every connected graph G with $\Delta = \Delta(G) \geq 2$ which is not exceptional satisfies*

$$\chi'_{\mathrm{rn}}(G) \leq 2\Delta - 1.$$

For $\Delta = 3$ this is the conjecture of [29] (for simple graphs). If you do not have access to the papers, contact us and we will send them to you. The work is split into three groups – group A tests the conjecture for graphs, group B for multigraphs, and group C studies regular graphs and looks for further graphs with $\chi'_{\mathrm{rn}}(G) \geq 2\Delta$.

Project 5: Rich-neighbor edge-colorings – A: testing the conjecture

1. Implement a program that computes $\chi'_{\mathrm{rn}}(G)$ (e.g. by ILP or backtracking). You may use the code from `https://github.com/anejrozman/Rich-neighbour-edge-coloring` or implement it yourself. Notice that this code was written for regular graphs (it takes the degree of the first vertex as Δ), so it has to be adapted for non-regular graphs; a SAT solver is often much faster than the ILP.

2. Test Conjecture 3 systematically on all connected graphs of maximum degree $\Delta = 3, 4, 5$ and small order (use `geng`), as far as it goes; here we focus on non-regular graphs (regular graphs are studied in group C). Which graphs attain the bound $2\Delta - 1$?

3. For larger graphs of maximum degree $\Delta \geq 4$, test the conjecture: search for a rich-neighbor $(2\Delta-1)$-edge-coloring by a random search or simulated annealing, and decide the graph exactly by ILP whenever the heuristic fails.

Report your results.

3.5 Graph indices

General Sombor index. The *general Sombor index* of a graph G is defined as

$$\mathrm{SO}_\alpha(G) = \sum_{v_iv_j \in E(G)} \left(d_G(v_i)^2 + d_G(v_j)^2\right)^\alpha,$$

where $d_G(v_i)$ is the degree of the vertex v_i, and $\alpha \in \mathbb{R}$. This invariant was recently studied in the class $\mathcal{T}_{n,\Delta}$ of trees on n vertices with maximum degree Δ. The following is known.

- *Maximum:* the extremal trees are known for every $\alpha \in \mathbb{R} \setminus [0,1]$ [6, 7]; for $\alpha \in (0,1)$ they are known for $\Delta \in \{3,4\}$ [7] and for $\lceil n/2 \rceil \leq \Delta \leq n-2$ [5].
- *Minimum:* the extremal trees are known for every $\alpha > 0$ [6]; for $\alpha < 0$ they are known for $\Delta \geq \lceil n/2 \rceil$ [6] and for $\Delta = 3$ [7]. Interestingly, for $\Delta = 3$ the structure of the minimal trees changes at $\alpha_0 \approx -4.57793$ [7].
- *Branching vertices:* the minimal trees on n vertices with a prescribed number b of branching vertices (vertices of degree at least 3) are known for $\alpha > 0$ [5].

If you do not have access to the papers, contact us and we will send them to you.
Trees of order n can be generated with `geng`: for example, `geng -c -D4 12 11:11` gives all trees of order 12 with maximum degree at most 4 (connected graphs with $n-1$ edges); in the groups with prescribed maximum degree Δ, discard the trees with smaller maximum degree. For larger n, `graphs.trees(n)` in Sage is faster; an exhaustive search is realistic up to about $n = 20$–22. Notice that $\mathrm{SO}_\alpha(T)$ depends only on the multiset of the pairs $\{d(u), d(v)\}$ over the edges uv, so non-isomorphic trees often have exactly the same value. Group the trees by this multiset first (this detects exact ties without numerical errors), and report the extremal trees as such classes, together with the number of non-isomorphic trees in each class (different classes can still tie for special values of α, e.g. $\alpha = 1$).

Project 6: General Sombor Index: Maximum Trees for $\alpha \in (0,1)$ and $\Delta = 5, 6$

Your task is to identify all trees with maximum $\mathrm{SO}_\alpha(G)$ among n-vertex trees with maximum degree Δ (consider separately $\Delta = 5$ and $\Delta = 6$) and $\alpha \in (0,1)$; the cases $\Delta = 3, 4$ are solved in [7]. In order to do so:

1. For each n starting from $2\Delta-1$ up to as large as computationally feasible, identify all trees with maximum $\mathrm{SO}_\alpha(G)$ among trees with maximum degree Δ (for $n \leq 2\Delta$ the answer is known [5], so these cases can be used to check your program). For $\Delta = 3, 4$, the extremal trees have at most one vertex of degree different from 1 and Δ [7]. Test whether this also holds for $\Delta = 5, 6$. Our computations suggest that it holds for all small n, so the main question is whether extremal trees with at least two such vertices exist for larger n; search for them by heuristics.

2. Take at least 10 different values of α from $(0,1)$, and perform a separate search for each of them. Include values very close to 0, very close to 1, and several intermediate points. For each n, compare how the extremal trees change when α changes.

3. For small n, perform a systematic search using `geng` to generate all trees with the given maximum degree. For larger n, use heuristic or metaheuristic methods, such as simulated annealing.

4. Optional: use AI tools to analyze the observed patterns and to argue why the extremal trees have the observed structure, suggesting partial proofs or structural insights.

Report your results.

3.6 Total geodesic distance

For a connected graph G, the *total geodesic distance* of G is

$$\mathrm{tgd}(G) = \sum_{\{u,v\}\subseteq V(G)} d_G(u,v)\,\sigma_G(u,v),$$

where $d_G(u,v)$ denotes the distance between u and v, and $\sigma_G(u,v)$ is the number of shortest paths (geodesics) between u and v in G. Thus $\mathrm{tgd}(G)$ is the total length of all geodesics of G. The invariant was introduced by Marjanović and Gutman [24], but remained largely overlooked, and it was recently revived in [3]. A graph is *geodetic* if $\sigma_G(u,v) = 1$ for all pairs $\{u,v\} \subseteq V(G)$. For every connected graph, $\mathrm{tgd}(G) \geq W(G)$, where $W(G)$ is the Wiener index, with equality if and only if G is geodetic. In particular, $\mathrm{tgd}(T) = W(T)$ for every tree T.

Let $H_{n,g}$ be the graph obtained from the cycle C_g by appending $n-g$ leaves to one vertex of the cycle, and for even g, let $L'_{n,g}$ be the graph obtained from $C_g = v_1v_2\cdots v_gv_1$ by appending a path of length $\lfloor (n-g)/2 \rfloor$ to v_1 and a path of length $\lceil (n-g)/2 \rceil$ to the antipodal vertex $v_{g/2+1}$. In [3] the extremal unicyclic graphs were determined: for odd girth g they coincide with those for the Wiener index, while for even girth g the maximum is attained only by $L'_{n,g}$ and the minimum only by $H_{n,g}$. Over all unicyclic graphs G on $n \geq 6$ vertices it holds that

$$\mathrm{tgd}(H_{n,3}) \leq \mathrm{tgd}(G) \leq \mathrm{tgd}(L'_{n,4}).$$

Among all connected graphs on n vertices, the minimum is attained by K_n, while the maximum is open (it is not the path P_n) [3]. If you do not have access to the papers, contact us and we will send them to you.

Hint for the groups in this section (and for the geodesic subpath number): do not enumerate paths. One breadth-first search from a vertex u gives the distances $d_G(u,v)$ and the numbers $\sigma_G(u,v)$ for all v (the number of shortest paths to v is the sum of these numbers over the neighbours of v one step closer to u), so $\mathrm{tgd}(G)$ can be computed in time $O(nm)$.

Project 7: Total geodesic distance: Cubic graphs

We consider connected cubic graphs on n vertices. The goal is to study the value of $\mathrm{tgd}(G)$ for cubic graphs and to investigate extremal structures.

We ask whether there is a structural reason that determines which cubic graphs have minimal or maximal values of $\mathrm{tgd}(G)$. In particular, it is interesting to investigate whether path-like cubic chains (for example obtained by concatenating copies of $K_4 - e$, with suitable end blocks) behave differently from highly symmetric cubic graphs. More generally, we ask whether there is any relation between $\mathrm{tgd}(G)$ and classical graph parameters such as diameter, girth, or the number of short cycles.

Your tasks are:

- Compute $\mathrm{tgd}(G)$ for all connected cubic graphs up to $n \leq 20$ (use `geng -c -d3 -D3 n`; it takes seconds for $n = 18$ and minutes for $n = 20$), and compare the extremal graphs with the extremal cubic graphs for the Wiener index (compute these too).

- Identify graphs that appear extremal with respect to $\mathrm{tgd}(G)$.
- Test whether any structural pattern emerges for small values of n.
- For larger values of n, where complete enumeration of all cubic graphs is not possible, use heuristic search methods. This means: start from an initial cubic graph and repeatedly modify it (for example by edge swaps or local rewiring), or generate random cubic graphs, and guide the search toward graphs with extreme values of $\mathrm{tgd}(G)$.
- For each tested graph G, compute and record additional structural parameters: the diameter of G, the girth of G, and basic cycle information (for example the number of 4-cycles and 6-cycles, and whether cycles overlap or form dense clusters). Then compare how these structural properties relate to the value of $\mathrm{tgd}(G)$, and look for patterns or correlations.
- Formulate a conjecture for the cubic graphs with minimum and maximum $\mathrm{tgd}(G)$, and try to prove it (possibly with the help of AI tools) for one special family, e.g. the chains of copies of $K_4 - e$.

Report your results.

3.7 Induced 2-regular subgraphs

A graph is 2-*regular* if all its vertices have degree 2, i.e. it is a disjoint union of cycles. For a set $S \subseteq V(G)$, let $G[S]$ denote the subgraph of G induced by S. For a cubic graph G, the *induced cycles vertex number* $c_{\mathrm{ind}}(G)$ is the largest size of a set S such that $G[S]$ is 2-regular. Henning, Joos, Löwenstein and Sasse [12] proved that $c_{\mathrm{ind}}(G) \geq (n+2)/4$ for every cubic graph G of order n, and proposed the following conjecture.

Conjecture 4. *If G is a cubic graph of order n, then $c_{\mathrm{ind}}(G) \geq n/2$.*

The conjecture holds for claw-free cubic graphs, where even $c_{\mathrm{ind}}(G) > 13n/20$ [12], and for connected 4-chordal cubic graphs other than K_4, $K_{3,3}$ and $K_2 \square K_3$, where $c_{\mathrm{ind}}(G) \geq 5n/8 + 3/4$ [11]. Another approach uses the $(1,2)$-domination number $\gamma_{1,2}(G)$, the smallest size of a set S such that every vertex outside S has a neighbor in S and every vertex of S has at least two neighbors in S: it is known that $n/2 \leq \gamma_{1,2}(G) \leq 3n/4$ for connected cubic graphs [10], so every cubic graph with $\gamma_{1,2}(G) \leq c_{\mathrm{ind}}(G)$ satisfies the conjecture. Erveš and Tepeh [9] conjectured that this inequality fails only for special graphs (the trilobites), but this was disproved in [18], where $c_{\mathrm{ind}}(G)$ and $\gamma_{1,2}(G)$ were computed for all connected cubic graphs of order $n \leq 22$. All these graphs satisfy Conjecture 4. The conjecture is tight: the Franklin graph ($n = 12$) has $c_{\mathrm{ind}}(G) = 6 = n/2$, so a counterexample must have $c_{\mathrm{ind}}(G)/n$ strictly below $1/2$. If you do not have access to the papers, contact us and we will send them to you.

Project 8: Induced 2-regular subgraphs of cubic graphs

1. Implement a program that computes $c_{\mathrm{ind}}(G)$ and a largest set S with $G[S]$ 2-regular, e.g. by the ILP with binary variables x_v and constraints $\sum_{u \in N(v)} x_u \geq 2x_v$ and $\sum_{u \in N(v)} x_u \leq 3 - x_v$, maximizing $\sum_v x_v$. You may write the code on your own, or use AI tools to help you design and write it. Check it on small cubic graphs, e.g. K_4, $K_{3,3}$, the prism, the Petersen graph and the Franklin graph.
2. Since all cubic graphs up to 22 vertices were already checked, test Conjecture 4 on larger graphs from special classes: cubic graphs of large girth (e.g. `geng -c -d3 -D3 -tf n` generates the cubic graphs of girth at least 5 directly, but it is slow and realistic only up to

about $n = 24$; for larger orders and larger girth use the lists from the House of Graphs), snarks, generalized Petersen graphs $P(n,k)$, and planar cubic graphs. Which graphs have the smallest value of $c_{\mathrm{ind}}(G) - n/2$?

3. For larger n (e.g. $24 \le n \le 60$), search for cubic graphs with a small value of $c_{\mathrm{ind}}(G) - n/2$ by simulated annealing over cubic graphs (e.g. with edge switches which keep the graph simple and connected, see the document *Computational tools for the projects*), hoping to find a counterexample. Since $c_{\mathrm{ind}}(G) - n/2$ takes only few integer values, add a secondary objective (e.g. a cheap parameter such as the number of short induced cycles). Report for each n the smallest value found and the graphs attaining it, and compare them with the trilobites and the Franklin graph.

4. Optionally, compute also $\gamma_{1,2}(G)$ for the graphs you find and compare it with $c_{\mathrm{ind}}(G)$.

Report your results.

AI declaration

In the preparation of this document, the AI assistant Claude (Anthropic, https://claude.ai) was used, in particular for drafting and editing some of the project descriptions and for computational checks of the conjectures stated in them. All content was reviewed by the authors, who take full responsibility for it.

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